

DSA 210 – Spring 2026 Project Proposal

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Title:

The Impact of Crude Oil Prices on Food Sector Stock Prices

Data Source and Collection Method

Two publicly available datasets will be downloaded from Kaggle. The first is the **Brent Oil Prices** dataset (kaggle.com/datasets/mabusalah/brent-oil-prices), which provides daily Brent crude oil closing prices. The second is the **S&P 500 Stock Data** dataset (kaggle.com/datasets/camnugent/sandp500), from which food sector companies — Coca-Cola (KO), Kraft Heinz (KHC), McDonald's (MCD), and Mondelez (MDLZ) — will be filtered. Both datasets will be merged on a daily date key, covering the 2000–2023 overlap period. Missing values due to weekends and holidays will be handled via forward-filling.

Characteristics of the Data

The Brent Oil Prices dataset contains approximately 8,000+ daily observations with two main columns: date and closing price (USD/barrel), spanning from 1987 to the present. The S&P 500 dataset contains millions of rows across all listed companies; the food sector subset used in this project will consist of roughly 20,000–25,000 daily observations across four companies, with columns for date, open, close, high, low, and volume. After merging, the final dataset will have approximately 6,000 rows (one per trading day, 2000–2023) and around 10 features.